Continuous Optimization: Problem Set 2

November 2, 2026

1. Recall the definitions of α-strong convexity and β-smoothness: for all x, y

$$\frac{\alpha}{2}\|y-x\|_2^2 \le f(y) - f(x) - \langle \nabla f(x), y-x\rangle \le \frac{\beta}{2}\|y-x\|_2^2.$$

We will examine equivalent 1-st and 2-nd order definitions of these properties. For this question, assume f is twice continuously differentiable.

1. Show if $f : \mathbb{R}^n \to \mathbb{R}$ is α-strongly convex or β-smooth, then the restriction $g(t) := f(x+tv)$ is $\alpha\|v\|_2^2$-strongly convex and $\beta\|v\|_2^2$-smooth for any $x, v \in \mathbb{R}^n$;
2. Show α-strong convexity and β-smoothness are equivalent to the conditions $\alpha I \preceq \nabla^2 f(x)$ and $\nabla^2 f(x) \preceq \beta I$, respectively. (Here the notation $A \succeq B$ means $A - B \succeq 0$ or equivalently $\forall v : \langle v, Av\rangle \ge \langle v, Bv\rangle$)
3. Show for an α-strongly convex function, we have distance bound on the optimizer

$$\|x - x^*\|_2 \lesssim \frac{\|\nabla f(x)\|_2}{\alpha}.$$

4. Show β-smoothness is equivalent to Lipschitz-ness of the *gradient*:

$$\forall x, y : \quad \|\nabla f(y) - \nabla f(x)\|_2 \le \beta\|y-x\|_2.$$

(You can use the following fact without proof: $0 \preceq H \preceq \beta I \iff \forall x : \|Hx\|_2 \le \beta\|x\|_2 \iff \forall x, y : |\langle x, Hy\rangle| \le \beta\|x\|_2\|y\|_2$)

Solution:

2. In the analysis of gradient descent for strongly convex and smooth functions we used the following crucial claim, known as the Polyak-Lojaseiwicz inequality (PL):

$$\|\nabla f(x)\|_2^2 \gtrsim \alpha(f(x) - f(x^*)),$$

where x^* is the optimizer.

1. Show that α-strong convexity implies this α-PL condition.
2. Show the same exponential convergence of gradient descent for α-strongly convex and β-smooth f, but replace the strong convexity condition with α-PL condition.
3. Find a function that satisfies $\alpha = 1$ PL condition but is not ε-strongly convex for any $\varepsilon > 0$. (Hint: consider convex quadratics.)
4. **Remark**: this condition even allows rigorous fast convergence analyses for certain non-convex functions!

Solution:

3. You can use the following lemma without proof in this question: for convex ℓ and convex differentiable r, let $f := \ell + r$ with $x_* = \arg\min f$. Then

$$f(x) = \ell(x) + r(x) \geq f(x_*) + D_r(x, x_*) = \ell(x_*) + r(x_*) + D_r(x, x_*).$$

1. Prove the Bregman Pythagorean result: for strictly convex $r : \mathbb{R}^n \to \mathbb{R}$, let $\Pi_K(x) := \arg\min_{z \in K} D_r(z, x)$, then

$$\forall z \in K : \quad D_r(z, x) \geq D_r(\Pi(x), x) + D_r(z, \Pi(x)).$$

2. Let $f(z) := D_r(z, x)$, then $D_f = D_r$, i.e.

$$D_f(z, y) = D_r(z, x) - D_r(y, x) - \langle \nabla r(y) - \nabla r(x), z - y \rangle = D_r(z, y).$$

Solution:

4. (Lap Chi Lau Course CS798, HW2 Q1) Suppose you are given an 'accelerated' algorithm $\mathcal{A}$ for minimizing strongly convex and smooth functions: given f that is α-strongly convex and β-smooth, the algorithm outputs x_{alg} satisfying $f(x_{\text{alg}}) - f(x^*) \leq \varepsilon$ in

$$T \lesssim \sqrt{\frac{\beta}{\alpha}} \cdot \log\left(\frac{f(x_0) - f(x^*)}{\varepsilon}\right)$$

iteration, where $x^* = \arg\min_x f(x)$ (unknown), and x_0 is the initial point.

- You would like to use algorithm $\mathcal{A}$, but apply it to f that is *not strongly convex*. Prove that you can still use $\mathcal{A}$ as a black box to give an 'accelerated' algorithm with the following guarantee: for convex f that is β-smooth and minimizer $x^* = \arg\min_x f(x)$, find output x such that $f(x) - f(x^*) \leq \varepsilon$ in

$$T \lesssim \sqrt{\frac{\beta \|x_0 - x^*\|_2^2}{\varepsilon}} \cdot \log\left(\frac{f(x_0) - f(x^*)}{\varepsilon}\right)$$

iterations, where again x_0 is the initial point. You are allowed to assume that you know the values $\|x_0 - x^*\|_2$ and $f(x_0) - f(x^*)$.

- Use this to derive the lower-bound for first-order methods for smooth convex functions via the lower-bound for smooth and strongly convex functions.

Solution: